Tone Relative to Substance in European ESG Disclosure: A Continuous Corpus-Relative Score, 2018–2024

1 Scope and contribution

Investors, lenders and supervisors read the ESG sections of corporate reports, and a recurring concern is that positive language may run ahead of the content that supports it. This paper measures that relation directly in the text: how the tone of a report's ESG disclosure stands relative to the substance of that disclosure. Tone is the polarity of the ESG text under the finance dictionary of Loughran and McDonald (2011); substance is the density, in the same text, of mentions of a documented ESG vocabulary (§3). We compute the measure for 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year over 2018–2024 (§2). All of these firms already had to publish a non-financial statement under the Non-Financial Reporting Directive from the first year of the window; what changed was how much of its content was prescribed, with the alignment indicators of the EU Taxonomy from 2022 (2023 for financial firms) and, for 2024, statements under the European Sustainability Reporting Standards (ESRS).

The measure is a continuous score, and it is relative to the corpus. Each component is standardised over the 343 reports, so a value locates a report within this corpus and has no meaning outside it; its sign says only whether the report lies above or below the corpus mean (§3.4). The idea behind the score, positive language in excess of the substance that supports it, fixes which direction of the score is of interest. The score describes disclosure, not conduct.

Two scores. We follow the architecture of Lagasio (2024): tone relative to substance, measured as a tone component minus a substance component, each rescaled across the sample. The main score, ESGSI (1), keeps that architecture and adapts its components to our corpus and question. The second score, the Extended ESGSI $\text{ESGSI}^{\text{ext}}$ (2), is the new metric this paper proposes. It adds quantification on the side of substance and unbacked commitments on the side of tone, with weights that are a choice rather than an estimate; we report how far the score moves when they change (§5).

Adapting the architecture. Lagasio (2024) introduced the index on a 2023 cross-section of 749 standalone English-language sustainability reports. Our question is how tone and substance evolve within the same firms over seven years, and four design choices follow from it.

- *Corpus and extraction.* A balanced panel needs a document that every firm publishes every year: for all but one of our firms, the annual report or the universal registration document. These filings embed ESG disclosure in a much larger financial and legal document, so we score the extracted ESG passages rather than whole reports (§3.1).
- *Substance as density.* Substance is the number of ESG mentions per 100 words of ESG text, an amount that is comparable across reports of very different length (§3.4). It measures how much of the text is about identifiable ESG matters, the density of ESG content, not whether that content is specific or verified.
- *Finance dictionary.* Tone is measured with the Loughran–McDonald dictionary, built for the language of corporate filings.

- *A continuous score.* Components are standardised as z-scores and the score is read as a position in the corpus, with no threshold (§3.4).

Because corpus, vocabulary and sentiment dictionary all differ, our scores are not numerically comparable with those of Lagasio (2024), and no such comparison is attempted.

Roadmap. §2 describes the corpus and §3 the extraction, vocabulary, scores and inference. §4 reports the scores over time, against the confound of reporting mandates, by firm group and by pillar; §5 tests their sensitivity to the choices of the method, and §6 states the limitations.

2 Data

To follow disclosure over time we need the same firms in every year, so that a change in the scores reflects a change in reporting rather than in the sample. The corpus consists of 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year from 2018 to 2024. The firms are the constituents of the EURO STOXX 50 in its 2024 composition that published a report for every year of the window; the one constituent without reports for 2018 and 2019 is excluded. For the firm formed by a merger in 2021, the reports for 2018 and 2019 are those of one of the merging companies and the report for 2020 that of the other, its legal predecessor, so its series changes issuer and document type before the merger. Requiring seven reports selects firms that were listed and reporting throughout, so the results describe large, continuously listed euro-area issuers. A report's year is the financial year it covers, and for the three firms whose financial year does not end in December, the year in which it ends. All documents are the English-language versions published by the firms. The panel is balanced: every firm contributes seven reports and every year 49. Of the 344 files collected, one 2019 report was present twice, as a re-issue with identical text, and the copy was removed by comparing document contents. For each firm-year we take one recurring annual document: an annual report (235 reports), a universal registration document (URD, 101) or, for one firm, a standalone sustainability report (7). Table 1 gives the corpus by country and document type. The document-type mix barely moves over time (12 URDs in 2018, 14 or 15 in every later year). Every measure is computed on the ESG zones extracted from each report, not on whole documents (§3.1). Firms enter the paper only through their country, their industry and anonymous positions.

Each firm carries one supersector of the Industry Classification Benchmark (ICB), assigned at firm level and not derived from the reports. The 49 firms fall into 16 supersectors and 10 of the 11 ICB industries: Financials (11 firms), Consumer Discretionary (10), Industrials (9), Consumer Staples (5), Health Care and Technology (3 each), and Basic Materials, Energy, Telecommunications and Utilities (2 each); Real Estate is absent.

Table 1: Corpus composition by country and document type. Industries: number of ICB industries represented among the country's firms. Every firm contributes one report in each year from 2018 to 2024.

Country	Firms	Industries	Annual report	URD	Sustainability report	Total
France	17	7	13	99	7	119
Germany	14	7	98	0	0	98
Netherlands	6	5	40	2	0	42
Italy	5	4	35	0	0	35
Spain	4	3	28	0	0	28
Finland	2	2	14	0	0	14
Belgium	1	1	7	0	0	7
Total	49	10	235	101	7	343

Country and document type. Document type is almost entirely determined by country: 99 of the 101 URDs are French and all 98 German reports are annual reports, which reflects national reporting practice rather than sampling. Within the pooled panel a country effect and a document-type effect can be told apart only through 15 documents (13 French annual reports and 2 Dutch URDs), and the seven standalone sustainability reports belong to a single French firm. We treat this as a property of the design: no document-type effect on either score is identified, and since type is constant within 43 of the 49 firms, firm fixed effects absorb nearly all of it (§3.5).

Industry. Industry cuts across countries: every industry has firms in at least two countries, and France and Germany each span seven of the ten, so industry contrasts can be drawn within a single country and document regime (§4.3). Industry cannot, however, be separated from the firm: industries hold 2 to 11 firms, industry is constant within each firm and so absorbed by firm fixed effects, and because the URD filers are 15 French firms and one Dutch firm, an industry's URD share is essentially its French share.

3 Method

We extract the ESG passages of each report (§3.1), count in them the mentions of a documented ESG vocabulary (§3.2) and combine tone, substance, quantification and the backing of commitments into two scores (§3.3); §3.4 explains the design and §3.5 the panel inference.

3.1 Extraction of ESG passages

The reports of §2 are annual reports and universal registration documents, in which ESG disclosure is embedded in a much larger financial and legal document. Scored whole, such a document would let financial-statement bulk and document length dominate every component. We therefore first extract its ESG passages; every later measure is computed on them alone.

Each PDF is read as text blocks in reading order, and filters keep only running prose. Page furniture is dropped: blocks entirely within the top or bottom 8 % of the page, and short margin blocks that recur, digits aside, on more than 30 % of a document's pages (running headers and footers). Tabular blocks (a high share of digits in a few lines, or short lines of which the first lacks terminal punctuation) are dropped unless they contain a vocabulary match. Short all-capital blocks are taken as section headings, which label the paragraphs that follow instead of entering the text. Navigation text (tables of contents, GRI and ESRS correspondence tables, stray page references), recognised by dotted leaders, trailing page numbers, multi-level section numbers and disclosure codes, is dropped, because such lists of ESG section names would otherwise pass the density rule below. Paragraphs are then stripped of web addresses, rejoined across line-break hyphenation and collapsed to single spaces.

Each paragraph is matched against the vocabulary of §3.2 on its unlemmatised text, ignoring case, with spaces and hyphens interchangeable between the words of a phrase and longer entries matched before the shorter ones they contain, so that *carbon footprint* counts once. A paragraph with at least one match per 100 words is retained together with one paragraph on each side; overlapping or adjacent windows are merged into a single zone, and zones shorter than 150 characters are discarded. The zones of a report, in document order, form its extracted text. Across the 343 reports, extraction kept 66,591 zones holding 28.3 million of the 71.4 million words of the PDF text layers, 39.7 % (per report, median 38.6 %, range 7.1–75.6 %). The share rises over the window, from a mean of 31.6 % of a report in 2018 to 49.6 % in 2024: ESG disclosure takes a growing part of the documents, although about a tenth of the rise comes from the transversal entries, umbrella terms and the names of reporting regimes (§6). The procedure depends only on the PDF and the vocabulary, and its recall is bounded by the vocabulary: a

passage that uses none of its entries is never selected. To size what is lost, we re-read 98 reports, two per firm and 14 per year, and classified passages of the retained and the discarded text with independent ESG classifiers (Schimanski et al., 2024). At most 8.1 % of the text that ESGBERT labels as ESG lies outside the extraction ([6.7, 9.8]; 13.0 % with FinBERT-ESG), with no trend over the window. This is an upper bound: in a reading of 40 discarded passages labelled as ESG, about half were audit reports, director biographies or product descriptions.

3.2 ESG vocabulary

The vocabulary defines what counts as an ESG mention, both for selecting passages and for measuring substance, so it is documented in full. Candidate entries came from three sources: the disclosure topics and terminology of reporting frameworks and standards (GRI, SASB, TCFD and the ESRS), including the names of the frameworks and regulatory regimes under which firms label their disclosure; the ESG vocabulary of dictionary-based studies of corporate disclosure; and a structured review of sample reports from the corpus, in which independent readers listed the ESG terms they found and independent evaluators judged the pooled list. **[Author: identify the reviewers of the candidate lists.]** A candidate was admitted if its occurrences in the extracted text, read in context, predominantly carry an ESG sense (a disclosure, target, data point or policy on an environmental, social or governance matter). Candidates from the review had in addition to be dispersed across firms, with at least 50 occurrences in the reports of at least 5 of the 49 firms. A polysemous head was admitted only inside enumerated collocations (*carbon footprint* rather than *footprint*; the ESG collocations of *social* and *water*), and a morphological family or a term carrying digits was written as a pattern (every word beginning *sustainab-* or *recycl-*; CO_2e; Scope 1–3; ISO standard numbers). Each entry belongs to one pillar: environmental (E), social (S), governance (G), or transversal (TRANS) for umbrella terms and reporting regimes (*sustainability*, *ESG*, *CSRD*). Entries whose occurrences concentrate in one industry are designated sectoral; they are part of the vocabulary, and excluding them is a sensitivity check (§5). Table 2 gives the composition.

Terms were admitted after reviewing a sample of their occurrences in context, and the frequency-weighted precision of the resulting vocabulary is estimated from those samples (3,401 occurrences of 430 entries), not measured directly on the final extraction, at about 82 % (95 % interval 80–84 %). The stability of the scores under perturbations of the vocabulary is reported in §5.

One vocabulary drives both extraction and scoring: all 434 entries select the passages, and the 402 terms, lemmatised, are what the substance component counts. Changing the vocabulary therefore changes the text on which every component is computed, not only the substance count. For scoring, the extracted text is lowercased, tokenised and lemmatised with a standard English

Table 2: Composition of the ESG vocabulary. Terms and collocations are literal phrases; a collocation stands in for a polysemous head. Sectoral entries cut across the three type columns. Scoring entries are the distinct lemmatised terms (patterns act at extraction only); the last column is their share of the 1,059,412 vocabulary occurrences in the extracted text of the 343 reports.

	All entries					Scoring entries	
Pillar	Terms	Colloc.	Patterns	Total	Sectoral	Entries	Occ. (%)
E (environmental)	123	29	16	168	14	152	38.2
S (social)	105	23	11	139	14	127	17.6
G (governance)	65	27	3	95	8	90	29.9
TRANS (transversal)	27	3	2	32	0	30	14.3
Total	320	82	32	434	36	399	100.0

language model; stopwords (the standard list and 189 corpus-specific words), punctuation, numerals and tokens shorter than three characters are removed, and the vocabulary passes through the same processing, so that each entry is matched in the representation of the text it is counted on. Eleven phrases that this processing would empty or reduce to a generic word (*well-being*, both of whose words are stopwords; *say on pay* to *pay*; *ISO 14001* to *iso*) are first replaced by single protected tokens, in text and vocabulary alike. The 402 terms yield $V = 399$ distinct lemmatised scoring entries.

3.3 Components and scores

Each score combines a measure of how positively a report speaks about ESG matters with measures of how much ESG content it provides. Let $d = 1, \ldots, n$ index the $n = 343$ reports and $j = 1, \ldots, V$ the scoring entries. Each report has two versions of its extracted text: the *lemmatised text* of §3.2, with L_d tokens, and the *raw text*, with case, digits and stopwords intact, with R_d tokens. Let c_{dj} be the number of occurrences of entry j in the lemmatised text of d, counted over word sequences of one to four tokens (the length of the longest entry). Entries are counted independently, so an occurrence of a multi-word entry also counts for any shorter entry it contains.

Substance. SUS is the density of ESG mentions, the number of vocabulary occurrences per 100 lemmatised tokens, with every entry weighted equally:

$$\mathrm{SUS}_d \;=\; \mathrm{SUS}_d^{\mathrm{dens}} \;=\; 100 \cdot \frac{\sum_j c_{dj}}{L_d}.$$

The superscript, used in tables and figures, marks the density specification as distinct from the alternatives examined in §5.

Tone. SEN is the Loughran–McDonald polarity (Loughran and McDonald, 2011) of the raw text,

$$\mathrm{SEN}_d \;=\; \frac{s_d^+ - s_d^-}{s_d^+ + s_d^-},$$

where s_d^+ and s_d^- count the words of the text, lowercased, that appear as written in the dictionary's positive (354 forms) and negative (2,355 forms) lists. The lists are lists of inflected forms and are matched as such, without stemming either side, which would admit words the dictionary does not contain (*objectives* or *diversity* as negative, *information* as positive). Following Loughran and McDonald (2011), a positive word preceded within three words by a negation (*no*, *not*, *none*, *neither*, *never*, *nobody*) is not counted. SEN ranges from -1 (only negative words) to 1 (only positive words) and depends on the balance of positive and negative words, not on how many there are.

Quantification. A report that discloses substance can also put numbers on it. QUANT counts such evidence on the raw text, because preprocessing removes numerals:

$$\mathrm{QUANT}_d \;=\; m_d / R_d,$$

where m_d is the number of matches of three pattern families: percentages other than zero; numbers of four or more consecutive digits that are magnitudes, excluding years, year ranges, standard codes (ISO 14001) and the numbers of legal acts; and units of mass, energy, power and volume preceded by a figure (*15 tonnes*, *3.5 MWh*). Only figures count: the names of reporting frameworks are not quantification and grow with reporting mandates; CO_2, *GHG* or a unit without a figure are topic words; and the zero cells of the taxonomy templates, which arrive with

the templates in 2022–2023, record no figure the firm chose to report. Numbers written with thousands separators are not counted either: in the extracted text fewer than a third of them sit next to an ESG metric, the rest belonging to financial and remuneration tables.

Unbacked commitments. A commitment that states no quantity, deadline or base year cannot be verified: a report that aims to reduce its emissions may mean 0.5 % or 50 %, by 2030 or by 2050 (Bingler et al., 2024). UNBACK measures how much of this backing the ESG commitments of a report lack. A sentence of the raw text, split within its paragraph, is an ESG commitment if it names the firm (*we*, *our*, *the Group*, *the Company*), contains a cue of commitment or aspiration (*commit to*, *aim to*, *our target*, *goal* or *ambition*, *pledge*, *plan*, *intend*, *seek* or *strive to*, *net zero*, *carbon neutral*, or *will* followed by a verb of action such as *reduce* or *achieve*) and at least one entry of the vocabulary of §3.2. Conditional sentences (*if*, *could*, *might*, *may*), cross-references, disclaimers on forward-looking statements and financial contexts (remuneration, assurance, dividends) are not commitments. Following the elements that the ESRS require of every target (European Commission, 2023), a commitment is backed by a quantity Q (a figure counted by QUANT, an amount of money, *halve* or *double*), a deadline D (*by*, *until* or *before* a year later than the report's) and a base year B (*compared with*, *versus* or *from* a year not later than the report's, or *baseline*), each sought in the sentence and in the sentences just before and after it in the same paragraph. Commitment k lacks the share $u_k = (3 - Q_k - D_k - B_k)/3$ of the three elements, and

$$\mathrm{UNBACK}_d \;=\; \frac{\sum_{k=1}^{n_d} u_k + n_0\,\bar{u}}{n_d + n_0},$$

the mean over the n_d commitments of the report, shrunk towards the mean $\bar{u}$ of all commitments in the corpus with $n_0 = 5$ pseudo-commitments, so that a report with few commitments does not take an extreme value (Table 3). The rules find 19,727 commitments, 43 per report at the median; 73 % carry none of the three elements, 18 % a quantity, 17 % a deadline and 5 % a base year. **[Author: precision of the commitment rules, from a reading of a random sample of the sentences they select.]** UNBACK measures how well what a report promises is backed, not how much it promises, and like QUANT it reads form, not truth: a commitment with all three elements may still not be met.

Pillars. For description only, the pillar density counts the mentions of a pillar's entries per 100 lemmatised tokens, so that the four pillar densities sum to SUS_d; for a set of reports, the pillar share pools the mentions of the set. Neither enters a score.

Scores. Each component x is rescaled across the corpus by a z-score, $Z(x)_d = (x_d - \bar{x})/\sigma_x$, with $\bar{x}$ the mean and σ_x the population standard deviation over all 343 reports pooled across

Table 3: How UNBACK scores sentences: four ESG commitments and a statement of past performance. Q, D, B: quantity, deadline and base year found in the sentence or its neighbours; u: share of the three that is missing. The report-level score is the mean of u over its commitments, shrunk towards the corpus mean.

Sentence	Q	D	B	u
We are committed to reducing our environmental footprint across our value chain.	–	–	–	1.00
Our ambition is to reach net zero emissions by 2050.	–	✓	–	0.67
We aim to cut Scope 1 and 2 emissions by 42 %.	✓	–	–	0.67
We commit to reducing absolute Scope 1 and 2 emissions by 42 % by 2030 compared with 2019.	✓	✓	✓	0.00
In 2023, we reduced our water consumption by 12 %.	not a commitment			

the seven years, so that yearly means of a score are comparable. The two scores are

$$\mathrm{ESGSI}_d = Z(\mathrm{SEN})_d - Z(\mathrm{SUS})_d, \tag{1}$$

$$\mathrm{ESGSI}^{\mathrm{ext}}_d = Z(\mathrm{SEN})_d - Z(\mathrm{SUS})_d - w_q\, Z(\mathrm{QUANT})_d + w_u\, Z(\mathrm{UNBACK})_d, \qquad w_q = w_u = 0.5. \tag{2}$$

ESGSI (1) is the main score of the paper: it follows the architecture of Lagasio (2024), tone minus substance with each rescaled across the sample, with the components defined above. A higher value means more positive tone relative to the ESG content of the text. The Extended ESGSI (2) is the new metric this paper proposes. Quantified disclosure is further evidence of substance, so Z(QUANT) is subtracted, on the same side as Z(SUS); a commitment without backing is a claim without substance, so Z(UNBACK) is added, on the side of tone. The score thus sets what a report claims (its tone and its unbacked commitments) against what it discloses (the density of its ESG content and its figures). The quantity element of UNBACK reuses the figure patterns of QUANT, so the two are correlated by construction (§3.4). The weights give each added component half the weight of an original one, so that together they weigh as much as one of them and the tone–substance contrast remains the core of the score. They are a stated choice, not an estimate, since no external criterion is available to calibrate them; §5 reports the scores over a grid of weights.

3.4 Design choices

The score contrasts two positions in the corpus: where a report stands on tone and where it stands on the density of its ESG content. It is a contrast of relative positions, not a residual. It does not model the tone that a report's substance would predict, and could not usefully do so here: the two components are only weakly correlated (0.20 pooled, 95 % interval −0.03 to 0.41 from resampling firms; 0.11 within firms and 0.25 between firm means), so tone net of substance is almost tone itself. Each component is chosen for that contrast.

Substance is an amount per unit of text. Counting ESG mentions per 100 words means that a long report does not score more substance merely for being long, so reports of very different length are comparable; and since the count uses no statistics of the other reports, a report's substance does not change when the sample does. As a check of convergent validity in the sense of Campbell and Fiske (1959), substance should move with quantification, which is counted with patterns on the raw text rather than with the vocabulary. Measures that keep the amount of disclosure agree with it to a similar degree (0.43 for density, 95 % interval 0.25–0.56 from resampling firms; 0.44 for a TF-IDF count divided by length), while a TF-IDF specification that scales each report's term vector to unit length, and so discards the amount, agrees much less (0.22). The agreement is mainly between firms (0.59 across firm means, 0.18 within firms), and it holds when the entries that both measures count are removed from the vocabulary (0.37). Since QUANT is also a component of $\mathrm{ESGSI}^{\mathrm{ext}}$, this is a check on substance, not on the score. §5 reports the scores under these alternative specifications. Tone is read with the Loughran–McDonald dictionary because it was built from corporate filings and classifies words by the sense they carry in that register (Loughran and McDonald, 2011).

Tone and substance come in different units, a bounded ratio and mentions per 100 words, so each is standardised before they are combined: a z-score expresses a report's value in standard deviations from the corpus mean, and the difference in (1) is then a difference of relative positions on a common scale. A score is measured in standard deviations of its components, not of itself: its own standard deviation is 1.27 for ESGSI and 1.61 for $\mathrm{ESGSI}^{\mathrm{ext}}$, and slopes are reported on the component scale. The scale has no natural zero. Zero is the corpus mean of the combination, and no external benchmark exists against which a boundary could be calibrated for these reports (§6). We therefore read each score as a relative position and apply no threshold: we compare positions within the corpus through differences of means, trends, rank correlations and the overlap of

the top and bottom deciles. The Extended ESGSI adds the two features of disclosure that the tone–substance contrast leaves out, whether substance is quantified and whether commitments are backed.

3.5 Inference on the panel

A firm's seven reports resemble each other from year to year, so treating them as 343 independent observations would overstate precision. We therefore estimate trends within firms and measure their uncertainty at the level of the firm. Write $d = (i, t)$ for the report of firm $i = 1, \ldots, M$ in year t, with $M = 49$. The trend is the slope of a score on calendar year with firm fixed effects, which compares each firm only with itself; in the balanced panel it equals the average of the firms' own slopes. Standard errors are clustered by firm, with the usual small-sample correction and a t distribution with $M - 1$ degrees of freedom: they allow any correlation among a firm's reports, but not a shock common to all firms in one year, which seven years are too few to account for, so the slope summarises the yearly means. Because 49 firms are a modest number of clusters, we also compute p-values by a wild cluster bootstrap (Cameron et al., 2008), which rebuilds the data under a zero trend by flipping the sign of all of a firm's residuals together ($+1$ or -1 with equal probability, $B = 9{,}999$ replications); the p-value is the share of replications whose clustered t statistic is at least as large in absolute value as the observed one.

For an industry or country, the slope is the average of its firms' slopes. With few firms, the clustered standard error of a group slope is too narrow, so we read it through the t interval on the group's M_k firm-level slopes ($M_k - 1$ degrees of freedom). Equality of slopes across groups is tested with a Wald test on the clustered variance, which overstates significance when a group has a single firm. Yearly means carry 95 % intervals from a regression on year indicators, clustered by firm.

4 Results

This section asks how tone relative to substance moved over 2018–2024 and why, how much of the movement reporting mandates could account for, and where firms, industries and countries stand. Components are in corpus standard deviations and ESGSI (1) and ESGSI^{ext} (2) in standard deviations of their components (§3.4), with trends and intervals estimated as in §3.5. Industries are ordered by mean ESGSI throughout.

4.1 The yearly series and the trend

Both scores fall in every year of the window: mean ESGSI from 0.557 in 2018 to -1.109 in 2024, mean ESGSI^{ext} from 0.871 to -1.037 (Table 4, Figure 1a). The within-firm trend is -0.239 per year for ESGSI (95% interval $[-0.303, -0.176]$) and -0.295 for ESGSI^{ext} ($[-0.374, -0.216]$), -0.189 and -0.184 standard deviations of each score itself, and both are significant under clustered and bootstrap inference alike ($p < 10^{-4}$). The decline of ESGSI is not linear. A model with a mean for every year fits better than the line ($p < 10^{-4}$), because of a break in 2024, when mean ESGSI drops by 0.89 against at most 0.27 in any earlier year. Over 2018–2023 the slope is -0.161 ($[-0.227, -0.095]$), still clearly negative, and the slope over the full window is a summary of a steady decline followed by a step. ESGSI^{ext} falls more evenly: its slope over 2018–2023 is -0.265 and linearity is not rejected ($p = 0.11$), because the decline of UNBACK before 2024 steepens the early years and its rise in 2024 softens the step (below). In both cases and the standard errors do not cover a shock common to all firms in one year (§3.5). The crossing of zero between 2021 and 2022 is not an event, since both scores average zero over the window by construction.

What moves steadily is substance. A lower score can mean less positive tone or more substance (Figure 1b). SUS^{dens} rises in every year, from 5.75 to 8.22 mentions per 100 tokens (0.384

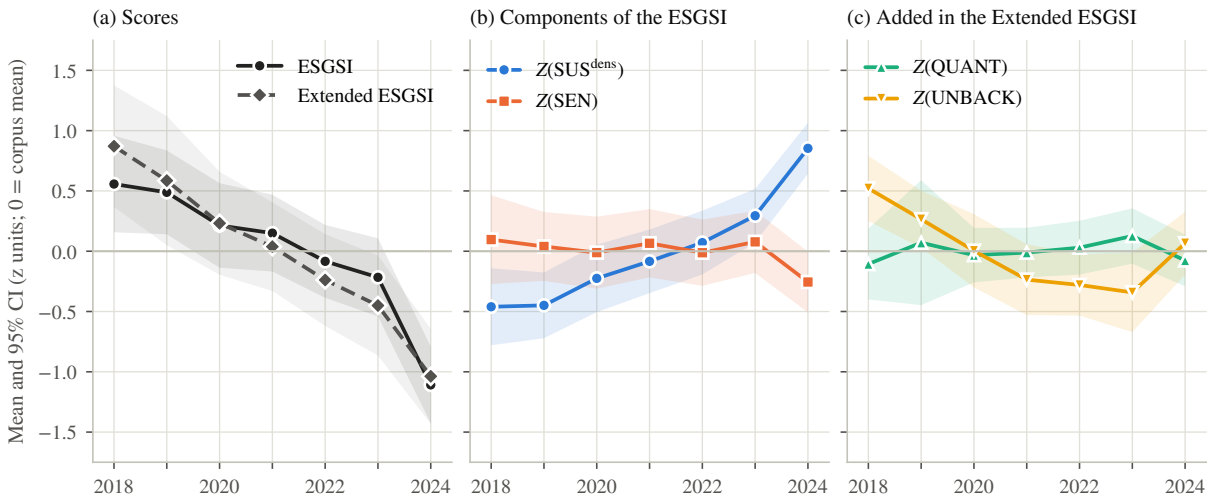


Figure 1: (a) Mean ESGSI and ESGSI^{ext} by year, 49 reports per year, with 95% intervals from year indicators and standard errors clustered by firm. (b) The same for $Z(\text{SUS}^{\text{dens}})$ and $Z(\text{SEN})$, whose difference is ESGSI (1). (c) The same for $Z(\text{QUANT})$ and $Z(\text{UNBACK})$, which the Extended ESGSI adds with weight one half (2). Components in corpus standard deviations, scores in standard deviations of their components; zero is the mean over the 343 reports.

Table 4: Yearly means and trend regressions; 343 reports of 49 firms, 49 per year. Panel A: means of the two scores and of their standardised components (1), (2); z-scores are taken over the pooled corpus (§3.3), so every row averages zero over the window. Panel B: slope on calendar year with firm fixed effects; standard errors and 95% intervals clustered by firm (t_{48}); p_{wild} from a wild cluster bootstrap with 9,999 replications (§3.5); R^2_w within firm. Component slopes are in corpus standard deviations per year, score slopes in standard deviations of the components (the standard deviation of ESGSI is 1.27, of ESGSI^{ext} 1.61).

A. Yearly means

	2018	2019	2020	2021	2022	2023	2024
ESGSI	0.557	0.488	0.214	0.150	−0.084	−0.217	−1.109
ESGSI^{ext}	0.871	0.586	0.232	0.038	−0.239	−0.450	−1.037
$Z(\text{SEN})$	0.097	0.039	−0.012	0.066	−0.012	0.078	−0.255
$Z(\text{SUS}^{\text{dens}})$	−0.461	−0.449	−0.225	−0.084	0.071	0.294	0.853
$Z(\text{QUANT})$	−0.106	0.071	−0.032	−0.011	0.030	0.126	−0.076
$Z(\text{UNBACK})$	0.521	0.265	0.004	−0.236	−0.280	−0.341	0.067

B. Trend

	Slope	SE_{cl}	95% interval		p_{cl}	p_{wild}	R^2_w
ESGSI	−0.239	0.032	−0.303	−0.176	1.1×10^{-9}	$< 10^{-4}$	0.349
ESGSI^{ext}	−0.295	0.039	−0.374	−0.216	1.2×10^{-9}	$< 10^{-4}$	0.373
$Z(\text{SEN})$	−0.035	0.026	−0.087	0.017	0.184	0.192	0.016
$Z(\text{SUS}^{\text{dens}})$	0.204	0.022	0.160	0.249	3.2×10^{-12}	$< 10^{-4}$	0.388
$Z(\text{QUANT})$	0.009	0.027	−0.045	0.064	0.729	0.765	0.001
$Z(\text{UNBACK})$	−0.102	0.028	−0.158	−0.046	6.4×10^{-4}	$< 10^{-3}$	0.071

a year, about 0.31 of its within-firm standard deviation), because mentions per report grow faster (+218%) than the extracted text that divides them (+127%). Tone has no trend over 2018–2023 (slope -0.005, $p = 0.87$); its yearly means stay within 0.11 of zero until 2024, when Z(SEN) falls to -0.255. That fall comes from the negative list, and from a particular part of it: the words that grow most between 2023 and 2024 are *negative*, *incidents*, *bribery*, *complaints*, *harassment*, *forced*, *corruption*, *violations* and *grievance*, the vocabulary of the negative impacts, incidents and grievance mechanisms on which the ESRS require firms to report (§4.2). With firm fixed effects the 2024 step of Z(SEN) is -0.30 ($[-0.50, -0.09]$) against 2018–2023; recomputing SEN without 24 negative words that name ESRS disclosure topics reduces it to -0.02 ($p = 0.87$), and a broader list of 48 such words gives the same result. The step sits mostly in fragments shorter than 30 words, the rows and labels of the ESRS data tables. This is not the whole of the change in tone in 2024: two independent classifiers also find the environmental and social narrative less positive, with fewer positive statements rather than more negative ones, a change that remains without those words and that the dictionary does not register (§5). Over the full window SEN has no detectable trend ($p = 0.18$); a failure to reject is not a demonstrated null. QUANT, counted as figures only, has no trend, but UNBACK falls by 0.102 a year (Figure 1c): the share of ESG commitments with no quantity, deadline or base year goes from 80 % in 2018 to 70 % in 2023, as targets come to carry a deadline (9 % of commitments in 2018, 20 % in 2021). In 2024 the reports make more commitments and back fewer of them, and Z(UNBACK) returns to the corpus mean. $\text{ESGSI}^{\text{ext}}$ therefore falls faster than ESGSI until 2023 and by about as much in 2024. In terms of disclosure practice, the reports devote a growing part of their ESG text to identifiable ESG content without putting more figures on it, back their commitments better until 2023, and in 2024 also report more adverse events: the scores fall because ESG disclosure became denser, its commitments more specific and, in the last year, its content more prescribed. The narrative of 2024 is also less positive, but the scores do not measure that change.

The panel is balanced, so the series is not a change in the set of firms or in the country mix, and the document-type mix barely moves (§2). Within Germany, fourteen firms with one country and one document type, both slopes are steeper than the pooled ones: -0.360 for ESGSI and -0.421 for $\text{ESGSI}^{\text{ext}}$ ($p < 0.001$ under both inferences). A trend of this sign therefore exists where country and document type do not vary, although this does not show that their mix plays no part in the pooled slope.

4.2 Regime vocabulary and the mandate confound

Every firm in the panel published a mandatory non-financial statement from 2018 onwards; over the window its content became increasingly prescribed, by the alignment indicators of the EU Taxonomy, first published for 2022 by non-financial firms and for 2023 by financial firms, and by statements under the ESRS for 2024. A firm that adds a taxonomy-alignment table or a standards-referenced sustainability statement adds ESG vocabulary without necessarily changing anything it does, and SUS^{dens} cannot tell the two apart. The vocabulary names regimes that did not exist or were not in force in 2018, and the transversal pillar that holds them nearly doubles its share of all mentions, from 10% to 19% (§4.4).

The purely definitional part can be bounded by deleting entries and rescoring, with the extraction held fixed. Deleting an explicit list of 22 regime, standard and rating names (3.9% of all mentions; deliberately over-inclusive, since several predate the window) leaves an ESGSI slope of -0.207 ($p < 0.001$), about six sevenths of the original, with a rank correlation of 0.994 with the full-vocabulary score. Deleting the whole transversal pillar (30 scoring entries holding 21 of the 22 names and 14.3% of all mentions) leaves -0.174 ($p < 0.001$ under both inferences), about three quarters, with a rank correlation of 0.973. $\text{ESGSI}^{\text{ext}}$ keeps -0.262 and -0.230 of its -0.295. Regime names thus account for about a seventh of the slope and the transversal pillar for about a quarter; the rest rests largely on environmental vocabulary that the same mandates also elicit.

Neither check disposes of the mandate confound. A regime elicits substantive disclosure as well as its own names (emissions, transition plans, most of the environmental vocabulary) in the words a firm that had changed its conduct would also use, and the environmental pillar is where the rise sits (§4.4). A mandate is a common time shock, collinear with calendar year; the balanced panel, the German subsample and the fixed effects remove cross-sectional composition, not anything that moves with time. The mandates offer only partial variation in exposure. Taxonomy alignment entered one year later for the financial firms, which report on a different template; we code it by the template each firm actually used, so that 35 firms first report alignment for 2022 and 14 for 2023. The ESRS were legally required for 2024 only where the corporate sustainability reporting directive had been transposed, for 22 firms, but 23 of the other 27 also published an ESRS-referenced statement for that year. With firm and year fixed effects, the first alignment year shifts $Z(\mathrm{SUS}^{\mathrm{dens}})$ by 0.05 (SE 0.13) and the legal ESRS requirement by -0.07 (SE 0.19); the smallest effects these designs could detect (80 % power) are 0.36 and 0.53 standard deviations, against a rise of 1.31 over the window. The steps in exposure produce no detectable jump of their own, which is consistent with anticipation, with a drift unrelated to the mandates, or with too little power, and the confound is left unresolved. Neither step moves ESGSI detectably either (-0.09, SE 0.17, and -0.07, SE 0.28). One consequence belongs to the architecture: since both scores fall when substance rises, a firm that adds a mandated disclosure moves down the scale for having complied. In 2024 the same holds on the side of tone, since reporting the negative impacts, incidents and grievances that the ESRS require adds words of the negative list (§4.1). A user of the scores should therefore read the decline over this window as denser disclosure, whatever its cause, and not as evidence that firms became more candid.

4.3 Firms, industries and countries

Most of what the scores record is a difference between firms. For ESGSI, 59.2% of the variation lies between firm means and 16.9% between year means, and the correlation between two reports of the same firm, net of year (the intraclass correlation), is 0.669 ($\mathrm{ESGSI}^{\mathrm{ext}}$: 63.7%, 13.8%, 0.700). The two components divide this differently: tone varies almost only between firms (69.7% of $Z(\mathrm{SEN})$ between firm means, 1.2% between years), while substance also moves with the year (56.9% and 18.4%). Positions persist from one year to the next and reshuffle over the window: the Spearman correlation of firms' ESGSI between consecutive years averages 0.842 (range 0.770 to 0.916) but is 0.303 between 2018 and 2024 ($\mathrm{ESGSI}^{\mathrm{ext}}$: 0.826 and 0.404). The decline is general but not uniform: 43 of the 49 firm slopes are negative for ESGSI and 46 for $\mathrm{ESGSI}^{\mathrm{ext}}$.

Industry and country are constant within firm, so every comparison between groups rests on 49 firms in cells of one to seventeen (Table 5), and every country contrast is also a document-type contrast (§2).

Levels. Industry means of ESGSI run from 1.085 (Technology, three firms) to -1.872 (Energy, two); Utilities has by far the highest $\mathrm{SUS}^{\mathrm{dens}}$ and the second-lowest score. Both ends rest on two or three firms, dispersion within an industry can exceed the distance between industries (Financials' firm means span -1.56 to 1.05), and the ordering moves within the window (Figure 2): Basic Materials and Telecommunications go from well above the corpus mean in 2018 to below it in 2024. $\mathrm{ESGSI}^{\mathrm{ext}}$ reorders the industries where quantification and the backing of commitments differ most: Energy, with the best-backed commitments, moves from -1.872 to -2.695 and Utilities, with the highest QUANT, from -0.980 to -1.977, while Technology, with the fewest figures, rises from 1.085 to 1.585. Within a single country and document regime, industry orders the firm means of ESGSI in France (adjusted R^2 of 0.557; 17 firms, seven industries) but not in Germany (-0.312; 14 firms, seven industries).

Germany stands apart among countries (0.611 against -0.056 for France), and in four of the five industries present in both countries the German mean exceeds the French one; the gap is

Table 5: Scores by ICB industry and by country, each ordered by mean ESGSI. Firms: number of firms, each contributing seven reports. Means are over the group's reports; SUS^{dens} in mentions per 100 tokens. Slope and 95% interval: group slope and t interval on the group's firm-level slopes (§3.5); none for a one-firm group.

			ESGSI			$\text{ESGSI}^{\text{ext}}$		
	Firms	SUS^{dens}	Mean	Slope	95% interval	Mean	Slope	95% interval
Industry (ICB)								
Technology	3	6.17	1.085	−0.345	[−0.748, 0.058]	1.585	−0.427	[−0.852, −0.002]
Telecommunications	2	6.12	0.841	−0.425	[−2.084, 1.235]	0.657	−0.593	[−3.220, 2.034]
Basic Materials	2	7.08	0.676	−0.429	[−0.921, 0.063]	0.302	−0.505	[−3.250, 2.241]
Consumer Discretionary	10	6.56	0.237	−0.356	[−0.536, −0.176]	0.624	−0.383	[−0.585, −0.180]
Industrials	9	6.41	0.160	−0.146	[−0.302, 0.010]	0.081	−0.232	[−0.374, −0.091]
Health Care	3	6.77	−0.154	−0.192	[−0.420, 0.036]	−0.117	−0.286	[−0.884, 0.313]
Consumer Staples	5	7.15	−0.229	−0.251	[−0.466, −0.035]	−0.860	−0.321	[−0.560, −0.082]
Financials	11	5.96	−0.254	−0.208	[−0.312, −0.104]	0.032	−0.278	[−0.377, −0.179]
Utilities	2	9.86	−0.980	0.295	[0.048, 0.543]	−1.977	0.513	[−2.086, 3.113]
Energy	2	7.33	−1.872	−0.295	[−1.338, 0.748]	−2.695	−0.291	[−3.081, 2.498]
Country								
Germany	14	5.63	0.611	−0.360	[−0.492, −0.229]	0.802	−0.421	[−0.556, −0.286]
France	17	6.85	−0.056	−0.233	[−0.305, −0.160]	−0.256	−0.328	[−0.427, −0.229]
Belgium	1	6.37	−0.109	−0.429	—	−0.733	−0.627	—
Finland	2	6.50	−0.157	−0.240	[−0.927, 0.447]	−0.434	−0.423	[−0.883, 0.038]
Spain	4	8.51	−0.194	−0.087	[−0.530, 0.357]	0.032	−0.078	[−0.504, 0.348]
Netherlands	6	7.04	−0.447	−0.222	[−0.500, 0.056]	−0.223	−0.251	[−0.513, 0.011]
Italy	5	6.65	−0.744	−0.029	[−0.329, 0.270]	−0.813	0.059	[−0.416, 0.534]

a contrast of country and document type at once, and neither is identified. By document type, annual reports average 0.023, URDs 0.021 and the one firm's sustainability reports −1.059. At report level, year explains 0.169 of the variance of ESGSI, industry adds 0.151 to 0.211 and country 0.060 to 0.120, depending on the order of entry, and firm identity still adds 0.309 of R^2 after year, industry, country and document type ($\text{ESGSI}^{\text{ext}}$: 0.276); across the 49 firm means, industry and country together reach an adjusted R^2 of 0.211 ($\text{ESGSI}^{\text{ext}}$: 0.339). Most of the difference between firms is firm-specific: a report's position says more about its firm's own reporting practice than about the firm's sector or country.

Slopes. Group slopes are read through the intervals on the group's firm-level slopes (Table 5, §3.5). Energy shows why: its ESGSI slope has a clustered interval of $[-0.415, -0.176]$ and a firm-slope interval of $[-1.338, 0.748]$. Read this way, the ESGSI decline is distinguishable from zero in Consumer Discretionary (10 firms), Consumer Staples (5), Financials (11), Germany (14) and France (17); Industrials (9) has the flattest decline among the larger industries, with an interval that just includes zero, and under $\text{ESGSI}^{\text{ext}}$ it also declines distinguishably. No group of one to three firms supports a statement about its slope: with two firms the interval rests on one degree of freedom, so even Utilities, whose two firms both rise, says nothing about the industry.

4.4 Pillar composition

The rise in mentions is not spread evenly across the pillars of §3.2 (Figure 3b). The environmental share of all mentions grows from 32% in 2018 to 42% in 2024 and the transversal share from 10% to 19%, while governance falls from 39% to 23% and the social share stays between 16% and 20%. In density, environmental mentions rise in every year, from 1.73 to 3.35 per 100 tokens, and transversal ones from 0.63 to 1.57, while governance mentions fall from 2.34 to 1.90: governance loses ground absolutely, not only by dilution. Two supervised classifiers reproduce the fall of the governance share (§5). The rise in substance is, in other words, environmental and regulatory



Figure 2: Mean (a) ESGSI and (b) $\text{ESGSI}^{\text{ext}}$ by ICB industry and year, in standard deviations of the components, on a diverging colour scale centred on the corpus mean. Industries are ordered by mean ESGSI; the number of firms, in parentheses, is also the number of reports averaged in each cell.

(§4.2).

Across industries the composition follows the business (Figure 3a). The environmental share runs from 28% (Financials) to 63% (Utilities), whose environmental density of 6.04 mentions per 100 tokens, against 4.11 in Energy, the next industry, gives it the highest SUS^{dens} in Table 5; Financials and Telecommunications are the governance-heavy industries (37% and 35%). The entries most over-represented in an industry are its own vocabulary (*hydroelectric* and *smart meter* in Utilities, *oil spill* and *methane* in Energy), so a firm scores substance partly for describing its business. The sectoral entries hold 1.0% of all mentions and at most 2.7% in any industry (Consumer Staples), too little to account for the position of Utilities or Energy. Mentions are concentrated (15 of the 399 scoring entries hold 48.1% of them); scoring without the sectoral entries, without each pillar and without the most frequent entry is examined in §5.

Reading the results. Neither score measures a quantity (§3.4): a yearly mean of −1.109 is a position within this corpus, and no level here is comparable with another application of the architecture. What is interpretable is the ordering within the sample and the direction of change: denser ESG disclosure, at unchanged dictionary tone until the prescribed statements of 2024, and firms differing far more than years. None of this is evidence about conduct: the reporting mandates of the same window could move the scores with conduct held fixed, and this design cannot exclude them (§4.2).

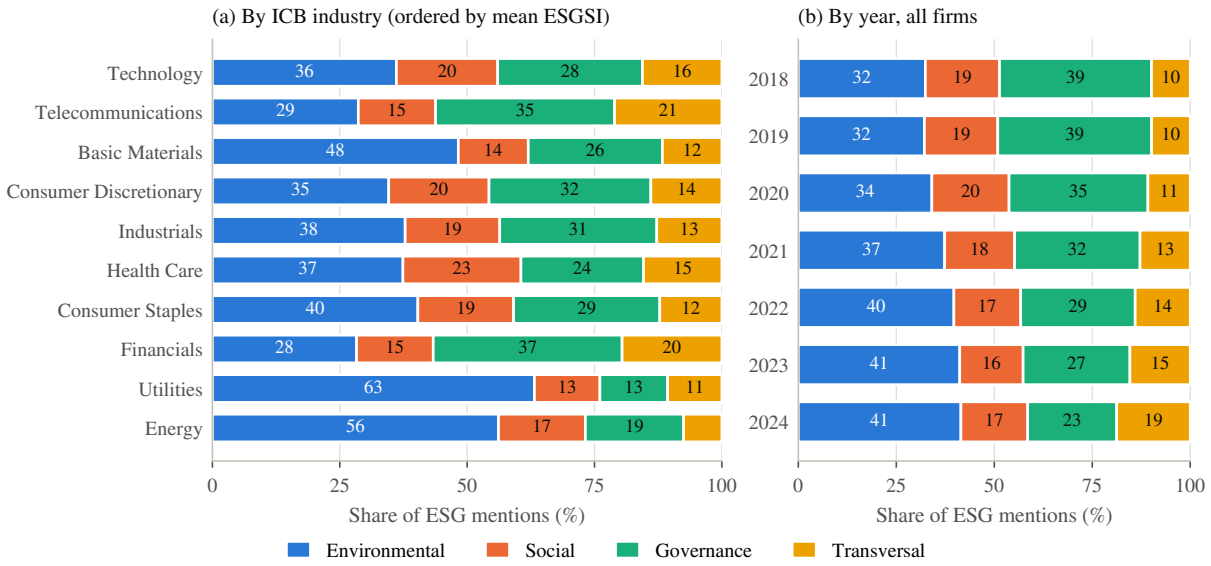


Figure 3: Share of all vocabulary mentions falling in each pillar, pooled over the group's reports, in percent: (a) by ICB industry, ordered by mean ESGSI; (b) by year, all 49 firms. The unlabelled segment (Energy, transversal) is 8%.

5 Robustness

A score built on a hand-made vocabulary and chosen weights is useful only if its conclusions do not hinge on those choices. We vary four of them—the substance specification, the weights of the Extended ESGSI, the vocabulary and the extraction rule—and compare each variant with a reference score over the same 343 reports. Every comparison asks three questions. Is the ranking of reports preserved (Spearman rank correlation ρ with the reference)? Are the same reports at the extremes (how many of the 34 reports with the highest, or lowest, reference score remain among the 34 highest, or lowest, written top/bottom)? And does the trend survive (the within-firm slope per year, §3.5)? The reference slopes are -0.239 for ESGSI and -0.295 for $\text{ESGSI}^{\text{ext}}$ (§4.1).

Substance specification. We recompute ESGSI with two TF-IDF versions of substance, in which each entry's count is weighted by how few reports use it. If the weighted count is divided by the length of the text, as density is, the ranking is almost unchanged ($\rho = 0.992$; 33/31 of the extreme deciles kept): how entries are weighted matters little. If instead each report's vector of weighted counts is scaled to unit length, the ranking changes markedly ($\rho = 0.626$; 13/19) and the slope weakens to -0.198: how the count is scaled matters much more (§3.4).

Weights of the Extended ESGSI. Over the grid $w_q, w_u \in \{0, 0.25, 0.5, 0.75, 1\}$ of Eq. (2), ρ with the score at the default weights is at least 0.871, the minimum being at $w_q = w_u = 0$, where the score is ESGSI, and at least 0.970 within 0.25 of the default on each axis, where at least 29 of the top and 29 of the bottom 34 reports remain. The ranking depends mostly on w_u, and so does the slope, since UNBACK has a trend of its own: it runs from -0.239 at $w_u = 0$ to between -0.342 and -0.351 at $w_u = 1$, and moves by at most 0.010 along w_q. It is negative in all 25 pairs, and every firm-clustered 95 % interval excludes zero.

Vocabulary. The vocabulary selects the text as well as scoring it (§3.2), so each deletion of entries is run in two ways: with the extraction held fixed, so that only the list counted by SUS changes, and with a simulated re-extraction, in which the extraction rule of §3.1 is re-applied

Table 6: Deletions from the vocabulary under simulated re-extraction ($n = 343$), against the full-vocabulary score as recomputed by the simulation (slope -0.239 for ESGSI and -0.295 for ESGSI^{ext}, as for the adopted scores). Entries: of the 434. Text: share of extracted words in paragraphs that still qualify (%). Top/bot: reports of the reference top/bottom 34 that remain. Slope: within-firm slope per year. Random rows: medians over 1,000 draws (text: mean). With caveat: six frequent entries admitted although part of their occurrences is navigation text, legal boilerplate or a non-ESG sense. Late admission: entries that would not pass the dispersion rule of §3.2 on the 2018–2019 reports alone. Late mass: entries with at least three quarters of their occurrences in 2022–2024.

			ESGSI			ESGSI^{ext}		
Deleted	Entries	Text	ρ	Top/bot	Slope	ρ	Top/bot	Slope
Sectoral entries	36	99.6	0.9983	33/32	−0.235	0.9986	34/33	−0.291
Pillar E	168	81.1	0.8883	27/25	−0.172	0.8704	28/26	−0.221
Pillar S	139	90.1	0.9718	31/29	−0.222	0.9818	30/33	−0.277
Pillar G	95	68.7	0.9048	21/28	−0.295	0.9514	26/29	−0.327
Pillar TRANS	32	94.8	0.9830	29/31	−0.183	0.9868	31/31	−0.238
Board of directors	1	93.6	0.9936	31/31	−0.248	0.9960	33/33	−0.303
With caveat	6	95.3	0.9902	30/31	−0.244	0.9912	31/33	−0.296
Late admission	100	99.5	0.9950	32/32	−0.202	0.9955	33/32	−0.257
Late mass	51	99.4	0.9879	31/31	−0.177	0.9903	33/30	−0.232
Random 10 %	43	96.2	0.9915	32/31	−0.234	0.9936	32/33	−0.289
Random 20 %	87	92.0	0.9804	30/30	−0.224	0.9845	31/32	−0.278
Random 30 %	130	86.8	0.9640	29/28	−0.212	0.9710	30/31	−0.265
Random 40 %	174	80.9	0.9434	27/27	−0.199	0.9541	28/29	−0.250
Random 50 %	217	73.8	0.9145	26/25	−0.176	0.9291	27/28	−0.225

without the deleted entries and all components are recomputed on the paragraphs that still qualify; with the full vocabulary the simulation reproduces the adopted scores ($\rho = 0.9999$). We delete designed sets of entries and, 1,000 times at each size, random sets of 10–50 % of the 434 entries, and read each designed deletion against 1,000 random deletions of the same size. Table 6 gives the re-extraction results for both scores, and Figure 4 those for ESGSI in both modes.

Removing the 36 sectoral entries barely moves either score ($\rho \geq 0.998$ in both modes): they carry about 1 % of the vocabulary's matches. The environmental pillar moves the score most. Without it $\rho = 0.888$, only 25 of the bottom 34 reports remain, and the slope falls to -0.172 (SE 0.036, $p < 0.001$; bootstrap $< 10^{-4}$), still clearly negative. Its ρ is lower than in 989 of the 1,000 random deletions of the same size: the environmental entries order the reports differently from the rest of the vocabulary. With pillar G, it is one of the two designed deletions whose ρ falls below the median for a random half of the vocabulary (Figure 4a). With the extraction held fixed its effect is larger ($\rho = 0.760$, 18 of the bottom 34 kept, slope -0.135 with $p < 0.001$), because the paragraphs extracted for environmental entries stay in the text, unscored, and dilute density. Without the transversal pillar the slope is -0.183; the remaining pillar and entry deletions leave it between -0.222 and -0.295, the steepest being without pillar G, which also removes nearly a third of the extracted text. ESGSI^{ext} moves less than ESGSI, in ρ, under every pillar deletion but the environmental one (0.870 against 0.888).

Two deletions target the concern that a vocabulary settled on the whole window favours the language of its last years. Deleting the 100 entries that would not have passed the dispersion rule on the 2018–2019 reports alone, or the 51 entries with at least three quarters of their occurrences in 2022–2024 (against 58 % of all occurrences), leaves ρ of 0.995 and 0.988 and slopes of -0.202 and -0.177: the trend does not rest on entries that only the later reports qualify.

Under random deletion the ranking degrades smoothly: with half of the entries removed the median ρ is 0.915 for ESGSI and 0.929 for ESGSI^{ext}, and 26 of the top and 25 of the bottom

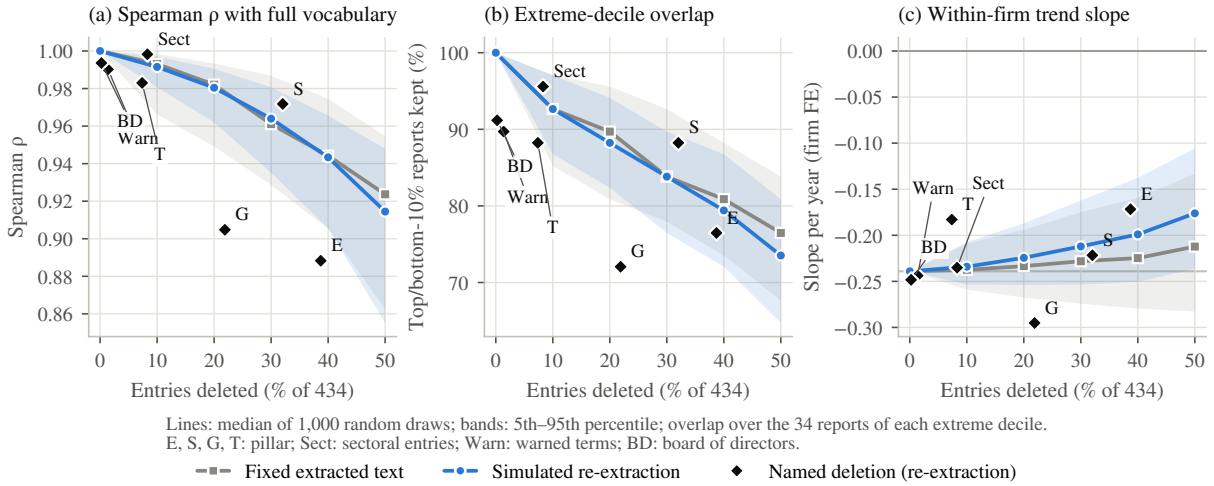


Figure 4: Stability of ESGSI under deletion of vocabulary entries ($n = 343$): random deletions in both modes, and the designed deletions of Table 6 under re-extraction, placed at the share of the 434 entries they remove (Sect: sectoral entries; T: pillar TRANS; Warn: entries with a caveat; BD: *board of directors*). (a) Spearman ρ with the full-vocabulary score. (b) Mean of the top- and bottom-decile overlaps. (c) Within-firm slope per year; the horizontal line is the full-vocabulary slope.

34 reports remain. The slope is the least stable measure. Under re-extraction its median drifts towards zero as more is deleted, from -0.234 at 10 % to -0.176 at 50 %, because less of the original text remains; with the extraction held fixed it stays between -0.238 and -0.212. Its sign does not change: over the 5,000 random draws the slope of both scores is negative in every draw, in both modes.

Extraction rule. The same simulation tightens the rule that selects the text, with the full vocabulary. A threshold of 1.5 or 2 matches per 100 words, or retaining hot paragraphs without their neighbours, keeps 70–93 % of the extracted text and gives ρ between 0.937 and 0.988 and slopes between -0.203 and -0.260. A looser rule than the adopted one cannot be simulated from the extracted text.

An independent text instrument. As a check of convergent and discriminant validity (Campbell and Fiske, 1959), we apply to the same extracted text, split into 318,258 paragraph-length passages, the open ClimateBERT classifiers (Webersinke et al., 2021) with which Bingler et al. (2024) build a cheap-talk index, the share of climate commitments that are not specific. Tone converges: the classifiers' net climate sentiment correlates with $Z(\text{SEN})$ at $\rho = 0.58$ (95 % interval $[0.42, 0.72]$ from resampling firms), within firms ($r = 0.51$) and between them ($r = 0.62$), and passage by passage (median within-report $\rho = 0.42$, positive in all but one report). Substance does not converge with specificity: SUS^{dens} tracks the share of climate passages ($\rho = 0.65$) but not the share of specific ones ($\rho = -0.11$; within firms $r = -0.35$), while QUANT does, weakly ($\rho = 0.23$). The cheap-talk index is unrelated to ESGSI ($\rho = -0.06$, $[-0.27, 0.14]$) and to $\text{ESGSI}^{\text{ext}}$ (0.03), so neither score is a proxy for cheap talk. UNBACK, which applies rules to a construct close to that of the index, converges with it weakly ($\rho = 0.18$, $[0.02, 0.33]$; within firms $r = 0.21$) and with the share of specific climate passages (-0.20), against a benchmark whose annotators agreed little on what is specific (Krippendorff's $\alpha = 0.17$; Bingler et al., 2024). It converges clearly with the share of climate passages that state a net-zero or reduction target according to a separate open classifier (Schimanski et al., 2023), applied to all 118,258 climate passages ($\rho = -0.50$, $[-0.58, -0.39]$; within firms $r = -0.47$): reports with more stated targets

have fewer unbacked commitments. In 2024 the classifiers also register a change, but a different one: fewer passages framed as opportunities and fewer commitments, not more framed as risks (net sentiment −0.69 standard deviations against 2018–2023). The check shares the text with the scores and covers climate passages only.

Classifiers of the whole text. Open classifiers that read all of the extracted text, applied to a random sample of 200 passages per report (67,448 passages), point the same way. Sentence-level tone from FinBERT (Huang et al., 2023) correlates with $Z(\text{SEN})$ at $\rho = 0.77$ ($[0.68, 0.84]$; within firms $r = 0.73$, between them 0.86), in environmental, social and governance passages alike (0.66, 0.69 and 0.41), and much less with every other component (at most 0.19, with $Z(\text{QUANT})$; −0.03 with $Z(\text{UNBACK})$, which is thus not a measure of tone). Pillar shares from FinBERT-ESG and ESGBERT (Schimanski et al., 2024) track ours ($\rho = 0.83$ for governance with both, 0.85–0.88 for the environmental and 0.69–0.70 for the social pillar), and both reproduce the fall of the governance share (from 0.45 to 0.29 and from 0.33 to 0.21 of ESG passages between 2018 and 2024). Substance again tracks how much ESG text a report holds (with the share of passages ESGBERT labels as ESG, $\rho = 0.72$), not how much of it describes concrete action (−0.03 with the share of environmental passages classified as actions). The classifiers also split the change in tone in 2024 into two parts. The fall of $Z(\text{SEN})$ is largest in fragments shorter than 30 words, the rows and labels of the ESRS tables (−0.47 standard deviations, and −0.10 without the ESRS topic words of §4.1). Separately, FinBERT's net tone falls by 0.27 standard deviations ($[-0.49, -0.06]$), in environmental and social passages, through fewer positive sentences and not more negative ones, and by as much without the passages dominated by those words. On the same passages without those words SEN does not move (+0.01): the dictionary misses a narrative that became less positive.

Except under unit-length scaling of substance, the ranking of reports is largely preserved and the trend stays negative; its size depends on the environmental and transversal vocabulary. These checks bound how far the scores depend on the choices varied, and the independent classifiers support the tone component, the pillar composition and, partly, the commitment component; they do not validate the scores as measures of conduct (§6).

6 Limitations

- *No external criterion.* Both scores are positions within this corpus and would move if the panel changed. Neither is tested against an external criterion (regulatory findings, controversy data, hand-coded cases), so no value is evidence about a firm's conduct; the empirical checks, convergent validity with QUANT (§3.4) and with independent classifiers (§5), read the same extracted text. The free external criteria that could be assembled for these firms (verified emissions under the EU Emissions Trading System, counts of reported controversies, market reactions) showed no detectable association and had power only for large effects; a test of consequences would need analyst forecasts and commercial controversy data.
- *A vocabulary built on this corpus.* Candidates, admission decisions, pillars and sectoral entries were settled on the 343 reports that are scored, with no part held out, and admission rests on the judgement of the reviewers of the candidate lists **[Author: identify the reviewers.]**. Its precision is estimated rather than measured on the final extraction (§3.2), and its recall is bounded only indirectly, by the share of the text that independent classifiers label as ESG and the extraction misses (§3.1). The stability of §5 is stability around this fit; it covers deletions of entries, including those that qualify only on the later reports, not additions, and the size of the trend depends on the environmental and transversal entries.
- *Selection by the scoring vocabulary.* The extraction selects text with the same vocabulary that measures substance (§3.1), so SUS^{dens} is measured on text chosen for its density of

the same entries, and the selection moves with the vocabulary. Entries that name reporting regimes (*taxonomy*, *ESRS*) admit the tables and statements those regimes prescribe, including tabular blocks that would otherwise be dropped. Part of the rise of the extracted share is therefore selection rather than more disclosure: without the transversal entries, which hold the regime names, the share would rise from 30.2 % of a report in 2018 to 46.4 % in 2024, against 31.6 % and 49.6 %, so about a tenth of the rise. Text admitted this way reaches every component: the topic words of the ESRS lower SEN in 2024 (§4.1), and the zero cells of the taxonomy templates would raise QUANT had they not been excluded (§3.3). The checks bound this dependence without removing it: re-extracting without the transversal pillar keeps most of the text and a clearly negative trend, deleting the entries that qualify only on the later reports leaves the trend (§5), and the share of classifier-labelled ESG text that the extraction misses shows no trend (§3.1).

- *Country, document type and industry.* Document type is almost determined by country (§2), so neither effect is identified on its own. Industries hold 2 to 11 firms and supersectors 1 to 7, both constant within firm, so group contrasts rest on few firms (§4.3).
- *Inference.* Standard errors are clustered on 49 firms and checked with a wild cluster bootstrap, but they do not cover shocks common to all firms in a year, and seven years are too few to cluster on (§3.5).
- *Mandate.* The content of the mandatory sustainability statement became progressively prescribed over the window. The staggered entry of taxonomy alignment and of the ESRS gives only weak variation in exposure, and it detects no jump (§4.2), so the rise in substance cannot be split into changed conduct and compliance, and compliance itself lowers both scores. The ESRS also require targets to state a base year and a target year, so part of any fall in UNBACK may be compliance as well.
- *Weights.* The weights of the Extended ESGSI are a choice. Its ranking is stable near them but depends on w_u further away, and so does the size of its trend, which is negative at every weight (§5).
- *Dictionaries.* SEN applies a finance dictionary to non-financial narrative, and as a bounded ratio it ignores how many sentiment words a text contains. Some negative words of the dictionary name ESG topics (*corruption*, *injury*), so a report that covers them more reads as more negative, and in 2024 it misses a fall in positive narrative tone that two classifiers find (§5).
- *Rules for commitments.* UNBACK rests on hand-written rules that match form, not sense: a deadline written in words (*within five years*) is missed, and a figure in a neighbouring sentence can back a commitment it does not refer to. It reads how well commitments are backed, not how many a report makes, and a report with few commitments is pulled towards the corpus mean (§3.3). Its quantity element reuses the figures that QUANT counts.

Data and code availability. The code, the derived data and a runbook that regenerates every number, table and figure are available in the project repository. The source reports are public documents published by the firms.

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